7 **(a)** In a nuclear fusion reaction, two low nucleon number (OR lighter) nuclei combine into a high nucleon number (OR heavier) nucleus.

(b) Since fusion occurs when the two nuclei touch each other, the distance at which the two nuclei fuse is $d = 2\times\left(1.2\times10^{-15}\right) = 2.4\times10^{-15}$ m .

Both nuclei must possess sufficient kinetic energy to overcome the electrostatic repulsion as they approach each other to fuse.

Minimum K.E. needed is when both nuclei just <u>come to rest</u> when they start to fuse.

By the principle of conservation of energy, as the two hydrogen nuclei approach each other,
decrease in K.E. = increase in E.P.E.

$$2\times E_{k,\text{min}} - 0 = \frac{e^2}{4\pi\varepsilon_0 d}$$

$$\begin{aligned}
E_{k,\text{min}} &= \frac{e^2}{8\pi\varepsilon_0 d} \\
&= \frac{\left(1.60\times10^{-19}\right)^2}{8\pi\left(8.85\times10^{-12}\right)\left(2.4\times10^{-15}\right)} \\
&= 4.7956\times10^{-14}\text{ J} \\
&= \frac{4.7956\times10^{-14}}{10^6\left(1.60\times10^{-19}\right)}\text{ MeV} \\
&= 0.299725\text{ MeV} = 0.30\text{ MeV} \quad \text{(shown)}
\end{aligned}$$

(c)

$$E_{k,\text{min}} = \frac{3}{2}kT$$

$$\begin{aligned}
T &= \frac{2E_{k,\text{min}}}{3k} \\
&= \frac{2\left(4.7956\times10^{-14}\right)}{3\left(1.38\times10^{-23}\right)} \\
&= 2.3167\times10^{9} = 2.32\times10^{9}\text{ K}
\end{aligned}$$

(d) **(i)** ${}^{0}_{1}\text{X}$

Particle X is a positron.

(ii) Since deuteron ${}^{2}_{1}\text{H}$ is more readily found, it suggests that reaction (2) is more probable than reaction (1) and that ${}^{2}_{1}\text{H}$ is <u>more stable</u> than ${}^{2}_{2}\text{He}$.
Hence, reaction (2) <u>releases more energy</u> than (1).

(e) Energy released,

$$
\begin{aligned}
E &= \Delta m c^2 \\
&= \left(\text{mass of reactants} - \text{mass of products}\right)c^2 \\
&= \left(2 \times 1.007825 - 2.014102 - 0.000549\right)\left(1.66 \times 10^{-27}\right) \times \left(3.00 \times 10^{8}\right)^2 \\
&= 1.4925 \times 10^{-13} = 1.49 \times 10^{-13} \text{ J}
\end{aligned}
$$

(f) Nuclear fission reaction of heavy nuclei requires much less energy to trigger (initiate) and can occur at a lower (room) temperature while the nuclear fusion reaction of light nuclei requires an extremely high temperature to trigger.

8 **(c)** This is to reduce collisions between protons and any gas molecules in the beam pipes.